Finite separation on subprobability powerdomains

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Abstract

We prove that the subprobability powerdomain of every FS-domain is an FS-domain. Starting from finitely separated Scott-continuous maps on the original domain, we construct monotone maps on simple subprobability valuations and a finite separator for each map. Extension from the simple-valuation basis and simultaneous approximation of finitely many way-below pairs then yield an FS approximate identity on the subprobability powerdomain.

Keywords: FS-domain; subprobability powerdomain; continuous valuation; stochastic order; finite separation.

1 Introduction

1.1 Domain theory and probabilistic approximation

Domain theory provides order-theoretic and topological foundations for denotational semantics and connects these structures with category theory. Directed suprema describe limits of approximations, Scott-continuous maps preserve these limits, and least fixed points on pointed dcpos interpret recursion. Cartesian closed categories provide products and function spaces for higher-order types [1, 2].

Continuous valuations represent probabilistic choice. The subprobability powerdomain $\mathcal{V}_{\leq 1}(D)$ consists of continuous valuations of total mass at most one, ordered pointwise on the Scott-open subsets of D. Although the assignment $D \mapsto \mathcal{V}_{\leq 1}(D)$ preserves continuity [3, Theorem 8], the category of all continuous dcpos is not Cartesian closed. The Jung–Tix problem asks for a category of domains that is closed under both function spaces and probabilistic powerdomains [3]. The full category of pointed FS-domains is Cartesian closed [1, Theorem 4.2.11]. It is therefore natural to ask whether D being an FS-domain implies that $\mathcal{V}_{\leq 1}(D)$ is again an FS-domain.

An FS-domain has a directed approximate identity of Scott-continuous maps finitely separated from the identity. For each such map $f : D \to D$, there is a finite $M \subseteq D$ such that

$$(\forall x \in D)(\exists a \in M) \qquad f(x) \leq a \leq x.$$

The image of a finitely separated map need not be finite. We use such a map and one of its finite separators to construct an FS approximate identity on the subprobability powerdomain. The construction does not require a least element in the original domain; the zero valuation is the least element of its subprobability powerdomain.

1.2 Outline of the proof

We begin with a finitely separated Scott-continuous map on the original domain. Its finite separator determines a finite poset, which is used to construct monotone maps on simple subprobability valuations. A rounding argument gives a fixed finite separator for each of these maps. We then extend the maps to all subprobability valuations and verify that they approximate any prescribed finite family of way-below pairs. A composition argument yields an FS approximate identity.

The proof uses finite-poset comparisons, rational-grid rounding, and self-composition, as in [4, Lemma 2.11, Proposition 3.13, and Lemma 4.4]. The finite-poset result of [4, Theorem 3.14] does not directly establish closure for arbitrary FS-domains: the approximating valuations need not be supported on a fixed finite subset of the original domain. Here the finite separator is used to construct finitely many separating valuations, without requiring the supports of the approximating valuations to lie in a single finite subset of the original domain.

2 Preliminaries

2.1 Domains and finite separation

Throughout the paper, dcpos are nonempty. A least element is assumed only when explicitly stated. A subset A of a poset is *directed* if it is nonempty and every pair of its elements has an upper bound in A. A *dcpo* is a poset in which every directed subset has a supremum. It is *pointed* if it has a least element. For a subset B of a poset, write

$$\uparrow B = \{x : (\exists b \in B)\ b \le x\}, \qquad \downarrow B = \{x : (\exists b \in B)\ x \le b\}.$$

We write $\uparrow x$ and $\downarrow x$ for singleton subsets.

For elements u, x of a dcpo, the relation $u \ll x$ means that, for every directed subset A,

$$x \le \sup A \quad \Longrightarrow \quad (\exists a \in A)\ u \le a.$$

A dcpo D is *continuous* if $\{u : u \ll x\}$ is directed with supremum x for each $x \in D$. A *domain* is a continuous dcpo. A subset $B \subseteq D$ is a basis if $B_x = \{b \in B : b \ll x\}$ is directed with supremum x for every x. We use interpolation in continuous dcpos:

$$u \ll x \quad \Longrightarrow \quad (\exists v)\ u \ll v \ll x.$$

The Scott topology $\sigma(D)$ consists of the upper subsets U such that $\sup A \in U$ implies $A \cap U \neq \varnothing$ for every directed A. A map between dcpos is Scott continuous precisely when it is monotone and preserves directed suprema. The dcpo $[D \to E]$ of Scott-continuous maps is ordered pointwise, and its directed suprema are computed pointwise. We write f^r for the r-fold composition of f, not for a pointwise power. All finite sums and unions may be empty, and $\mathbb{N} = \{0, 1, 2, \ldots\}$. For $r \in \mathbb{R}$, write $r^+ = \max\{r, 0\}$. For a subset B of a set X, the indicator $\mathbf{1}_B$ has value one on B and zero outside B.

Definition 2.1. A Scott-continuous map $f : D \to D$ is *finitely separated from the identity* if some finite set $M \subseteq D$ satisfies

$$(\forall x \in D)(\exists a \in M) \qquad f(x) \le a \le x. \tag{2.1}$$

Such an M is called a finite separator for f, as in [4, Definition 2.4]. An *FS approximate identity* is a family $(f_i)_{i\in I}$ indexed by a nonempty directed poset I, where each f_i is Scott continuous and finitely separated, $i \le j$ implies $f_i \le f_j$, and $\sup_i f_i = \mathrm{id}_D$ pointwise. A dcpo admitting an FS approximate identity is an *FS-domain*. Equivalently, one may use a directed subset of $[D \to D]$ with these properties, indexed by its pointwise order.

We use the following classical facts without reproving them.

Lemma 2.2. *[2, Lemma II-2.16] If $f : D \to D$ is Scott continuous and finitely separated, then $f(x) \ll x$ for every $x \in D$. Every FS-domain is continuous.*

Neither assertion of Lemma 2.2 requires a least element in D. In particular, a finitely separated map satisfies $f \le \mathrm{id}_D$, but its image need not be finite.

2.2 Continuous valuations

Equip $[0,\infty]$ with its usual order and Scott topology. A *continuous valuation* on a dcpo D is a map $\nu : \sigma(D) \to [0,\infty]$ that is strict, monotone and modular, and preserves directed unions. Thus

$$\nu(\varnothing) = 0, \qquad \nu(U) + \nu(V) = \nu(U \cup V) + \nu(U \cap V),$$

and

$$\nu\Big(\bigcup_d U_d\Big) = \sup_d \nu(U_d) \quad \text{for every directed family } (U_d) \subseteq \sigma(D).$$

Write $\mathcal{V}(D)$ for the dcpo of continuous valuations and

$$\mathcal{V}_{\le 1}(D) = \{\nu \in \mathcal{V}(D) : \nu(D) \le 1\}.$$

Both are ordered pointwise on Scott-open sets, and directed suprema are computed pointwise. The zero valuation is the least element of both $\mathcal{V}(D)$ and $\mathcal{V}_{\le 1}(D)$. Modularity implies finite subadditivity:

$$\nu\Big(\bigcup_{j=1}^{r} U_j\Big) \le \sum_{j=1}^{r} \nu(U_j). \tag{2.2}$$

For $x \in D$, the Dirac valuation is $\delta_x(U) = \mathbf{1}_U(x)$. A *simple subprobability valuation* is a finite sum $\sum_j a_j \delta_{x_j}$ with $a_j \ge 0$ and $\sum_j a_j \le 1$. Write $\mathcal{B}(D)$ for the set of simple subprobability valuations. After combining equal points, write $\alpha = \sum_{x\in X} \alpha_x \delta_x$ with the points of X distinct and $\alpha_x > 0$ for every $x \in X$. We set $\mathrm{supp}(\alpha) = X$; this notation denotes the finite set of points with positive coefficients, not a topological support. The paragraph after Lemma 2.4 proves that X and the coefficients α_x do not depend on the chosen finite Dirac-sum expression. The zero valuation has empty support. Finite sums and nonnegative scalar multiples of valuations are defined pointwise on opens.

Lemma 2.3. *[3, Theorem 8] If D is continuous, then $\mathcal{V}_{\le 1}(D)$ is continuous and, for every $\nu \in \mathcal{V}_{\le 1}(D)$, the set*

$$\{\alpha \in \mathcal{B}(D) : \alpha \ll \nu\}$$

is directed with supremum ν.

The hypothesis is continuity of D, not pointedness. The zero valuation belongs to $\mathcal{B}(D)$ for every D, since the empty sum is allowed.

2.3 Finite-poset comparisons

We use the coordinate notation for valuations on finite posets from [4, Section 2.3]. For a finite poset P, let $\mathrm{Up}(P)$ be the family of its upper sets, and put

$$\Delta_{\le 1}(P) = \left\{ p \in [0,\infty)^P : \sum_{x\in P} p_x \le 1 \right\}.$$

Every valuation of finite total mass on P has a unique expression $\sum_{x\in P} p_x \delta_x$, where

$$p_x = \nu(\uparrow_P x) - \nu(\uparrow_P x \setminus \{x\}).$$

Thus $\mathcal{V}_{\le 1}(P)$ is identified with $\Delta_{\le 1}(P)$ as an ordered set. For $A \subseteq P$, write $p(A) = \sum_{x\in A} p_x$. For arbitrary nonnegative vectors of finite total mass, the stochastic order is

$$p \le_{\mathrm{st}} q \quad \Longleftrightarrow \quad p(U) \le q(U) \quad (U \in \mathrm{Up}(P)).$$

This order is generally weaker than coordinatewise comparison of p_x and q_x. For an integer $N \ge 1$, define

$$M_N(P) = \{ p \in \Delta_{\le 1}(P) : Np_x \in \mathbb{N} \text{ for every } x \in P \}. \tag{2.3}$$

The set $M_N(P)$ is finite, because each of its coordinates belongs to $\{0, 1/N, \dots, 1\}$. The next lemma is [4, Lemma 2.11]. It characterizes the order between two finite sums of Dirac valuations with a common finite set of points and derives the corresponding inequality for nonnegative monotone functions.

Lemma 2.4. *Let D be a dcpo, $X \subseteq D$ finite, and*

$$\alpha = \sum_{x\in X} p_x \delta_x, \qquad \beta = \sum_{x\in X} q_x \delta_x,$$

where the coefficients are finite nonnegative real numbers. Then $\alpha \le \beta$ if and only if $p \le_{\mathrm{st}} q$ on the induced finite poset X. If these conditions hold, then every nonnegative monotone function $w : X \to \mathbb{R}$ satisfies

$$\sum_{x\in X} p_x w(x) \le \sum_{x\in X} q_x w(x). \tag{2.4}$$

Proof. Let $V \subseteq X$ be upper in the induced order and define

$$U_V = D \setminus {\downarrow_D}(X \setminus V).$$

Since $X \setminus V$ is finite and each principal lower set is Scott closed, $U_V \in \sigma(D)$. If $x \in X \setminus V$, then $x \notin U_V$. If $x \in V$ and $x \notin U_V$, there is $y \in X \setminus V$ with $x \le y$, contradicting upperness of V. Hence $U_V \cap X = V$. Consequently, $\alpha \le \beta$ implies

$$\sum_{x\in V} p_x = \alpha(U_V) \le \beta(U_V) = \sum_{x\in V} q_x.$$

Conversely, $U \cap X$ is upper in X for every $U \in \sigma(D)$. Thus $p \le_{\mathrm{st}} q$ implies $\alpha(U) \le \beta(U)$ for all $U \in \sigma(D)$, and therefore $\alpha \le \beta$.

Suppose $p \leq_{\mathrm{st}} q$. Let $0 < c_1 < \cdots < c_r$ be the positive values of w, and put $c_0 = 0$ and $V_j = \{x \in X : w(x) \geq c_j\}$. Since w is monotone, each V_j is upper. Because

$$w = \sum_{j=1}^{r} (c_j - c_{j-1}) \mathbf{1}_{V_j},$$

we obtain

$$\sum_x p_x w(x) = \sum_{j=1}^{r} (c_j - c_{j-1}) \sum_{x \in V_j} p_x \leq \sum_{j=1}^{r} (c_j - c_{j-1}) \sum_{x \in V_j} q_x = \sum_x q_x w(x).$$

If $w = 0$ or $X = \varnothing$, both sides of (2.4) are zero. □

Suppose that one simple valuation is written both as $\sum_{x \in X} p_x \delta_x$ and as $\sum_{y \in Y} q_y \delta_y$. Put $Z = X \cup Y$ and extend both coefficient families to Z by assigning coefficient zero to every point absent from the corresponding sum. Lemma 2.4, applied in both directions, gives

$$\sum_{z \in V} p_z = \sum_{z \in V} q_z \qquad (V \subseteq Z \text{ upper in the induced order}).$$

For each $z \in Z$, both $\uparrow_Z z$ and $\uparrow_Z z \setminus \{z\}$ are upper in Z, and hence

$$p_z = \sum_{y \in \uparrow_Z z} p_y - \sum_{y \in \uparrow_Z z \setminus \{z\}} p_y = \sum_{y \in \uparrow_Z z} q_y - \sum_{y \in \uparrow_Z z \setminus \{z\}} q_y = q_z.$$

Thus the points with positive coefficients and their coefficients are uniquely determined by the valuation. Every formula in this paper that uses these coefficients is therefore independent of the chosen finite Dirac-sum expression.

2.4 Extension and finite separation

The next lemma extends monotone maps from a basis to the whole dcpo. It also shows that a finite separator for the map on the basis remains a finite separator after extension. We will apply this result to maps first defined on simple valuations.

Lemma 2.5. *Let X be a continuous dcpo with basis B, and let $q : B \to X$ be monotone. Then*

$$\widehat{q}(x) = \sup\{q(b) : b \in B,\ b \ll x\} \tag{2.5}$$

is well defined and Scott continuous. Moreover, $b \in B$ and $b \ll x$ imply $q(b) \leq \widehat{q}(x)$. If a finite set $\Gamma \subseteq X$ satisfies

$$(\forall b \in B)(\exists \gamma \in \Gamma) \qquad q(b) \leq \gamma \leq b, \tag{2.6}$$

then the same set Γ separates $\widehat{q}$ from id_X.

Proof. For $x \in X$, let $B_x = \{b \in B : b \ll x\}$. Since B_x is nonempty and directed and q is monotone, $q[B_x]$ is directed. Hence $\widehat{q}(x)$ exists. Since $x \leq y$ implies $B_x \subseteq B_y$, the map $\widehat{q}$ is monotone.

Let $x = \sup_{d \in J} x_d$, where $(x_d)_{d \in J}$ is directed. If $b \in B_x$, choose $b \ll z \ll x$ by interpolation and then $d \in J$ with $z \leq x_d$. Thus $b \ll x_d$. Conversely, $b \ll x_d \leq x$ implies $b \ll x$. Therefore

$$B_x = \bigcup_{d \in J} B_{x_d}, \qquad \widehat{q}(x) = \sup_{d \in J} \widehat{q}(x_d).$$

Hence $\widehat{q}$ is Scott continuous. If $b \in B$ and $b \ll x$, then $q(b) \leq \widehat{q}(x)$ by the defining supremum.

Assume (2.6) and fix $x \in X$. Put $\Gamma_x = \Gamma \cap {\downarrow}x$. Since each $b \in B_x$ has a witness $q(b) \leq \gamma \leq b \leq x$,

$$q[B_x] \subseteq \bigcup_{\gamma \in \Gamma_x} {\downarrow}\gamma.$$

The set on the right is Scott closed: each ${\downarrow}\gamma$ is Scott closed and Γ_x is finite. Since $q[B_x]$ is directed,

$$\widehat{q}(x) = \sup q[B_x] \in \bigcup_{\gamma \in \Gamma_x} {\downarrow}\gamma.$$

Consequently $\widehat{q}(x) \leq \gamma \leq x$ for some $\gamma \in \Gamma_x$. Thus Γ separates $\widehat{q}$. □

For $b \in B$, one has $\widehat{q}(b) \leq q(b)$, but equality is not assumed. The implication used below is $b \in B$, $b \ll x \Longrightarrow q(b) \leq \widehat{q}(x)$.

The next proposition combines Lemma 2.5 with self-composition. Lemma 4.4 of [4] compares the self-compositions of maps drawn from two increasing approximation families. The proposition below instead assumes condition (ii), which requires one map to approximate any prescribed finite list of pairs $u_j \ll b_j$ on the basis. It does not assume that the maps in $\mathcal{A}$ form a directed family.

Proposition 2.6. *Let X be a continuous dcpo with basis B. Then X is an FS-domain if and only if there is a nonempty family $\mathcal{A}$ of monotone maps $q : B \to X$ satisfying:*

(i) *for each $q \in \mathcal{A}$, there is a finite $\Gamma_q \subseteq X$ such that*

$$(\forall b \in B)(\exists \gamma \in \Gamma_q) \qquad q(b) \leq \gamma \leq b;$$

(ii) *for every finite list $u_j \ll b_j$, with $b_j \in B$, there is one $q \in \mathcal{A}$ such that*

$$u_j \leq q(b_j) \qquad \text{for every } j.$$

For any such family, define $\widehat{q}(x) = \sup_{b \in B,\, b \ll x} q(b)$. Then $\{\widehat{q}^{\,2} : q \in \mathcal{A}\}$ is an FS approximate identity on X, ordered pointwise.

Proof. Suppose first that $(f_i)_{i \in I}$ is an FS approximate identity on X. Take $\mathcal{A} = \{f_i|_B : i \in I\}$. Each restriction has the same finite separator as f_i, so (i) holds. For a finite list $u_j \ll b_j$, the equality $b_j = \sup_i f_i(b_j)$ gives an index i_j with $u_j \leq f_{i_j}(b_j)$. A common upper bound of the finitely many i_j gives (ii). For an empty list any index suffices.

Conversely, suppose (i)–(ii) hold. By Lemma 2.5, each $\widehat{q}$ is Scott continuous and is finitely separated by Γ_q. Put $\mathcal{H} = \{\widehat{q} : q \in \mathcal{A}\}$.

First, let $u_j \ll x_j$ be a finite list in X. Choose $u_j \ll v_j \ll x_j$. Since B_{x_j} is directed with supremum x_j, choose $b_j \in B_{x_j}$ with $v_j \leq b_j$. Then $u_j \ll b_j \ll x_j$. Condition (ii) gives a common $q \in \mathcal{A}$ such that

$$u_j \leq q(b_j) \leq \widehat{q}(x_j) \qquad \text{for every } j. \tag{2.7}$$

Next, let $p_1, \ldots, p_r \in \mathcal{H}$ and choose a finite separator M_j for each p_j. For every $a \in M_j$, Lemma 2.2 and interpolation give

$$p_j(a) \ll z_{j,a} \ll a.$$

Apply (2.7) to the finite list of all pairs $(p_j(a), z_{j,a})$ and $(z_{j,a}, a)$. There is $h \in \mathcal{H}$ such that

$$p_j(a) \le h(z_{j,a}), \qquad z_{j,a} \le h(a) \qquad (a \in M_j,\ 1 \le j \le r).$$

For $x \in X$, choose $a \in M_j$ with $p_j(x) \le a \le x$. Since p_j and h are monotone,

$$p_j^2(x) \le p_j(a) \le h(z_{j,a}) \le h^2(a) \le h^2(x).$$

Thus $\{h^2 : h \in \mathcal{H}\}$ is directed. Since $h^2 \le h$, every finite separator for h also separates h^2.

Finally, fix $u \ll x$ and choose $u \ll z \ll x$. Apply (2.7) to (u, z) and (z, x). The resulting $h \in \mathcal{H}$ satisfies

$$u \le h(z) \le h^2(x).$$

Since X is continuous and $h^2 \le \mathrm{id}_X$,

$$x = \sup_{u \ll x} u \le \sup_{h \in \mathcal{H}} h^2(x) \le x.$$

Therefore $\{h \circ h : h \in \mathcal{H}\}$ is an FS approximate identity. □

3 Maps on simple valuations

This section constructs monotone maps on simple subprobability valuations from a finitely separated Scott-continuous map on the original dcpo. A finite poset associated with the given separator is used to define the maps and compare them when the input valuation increases. We then prove that each map is separated from the identity by a finite set of simple valuations. The extension to arbitrary subprobability valuations is addressed in the next section.

3.1 A finite poset from finite separation

Set

$$\pi(x) = \pi_{f,M}(x) = \{a \in M : f(x) \le a\}, \qquad P = \pi[D]. \tag{3.1}$$

Order P by reverse inclusion: $u \le v$ means $u \supseteq v$. Every $u \in P$ is nonempty, and π is monotone. Choose

$$a_u \in f[u] \cap \bigcap_{a \in u} {\downarrow}a \qquad (u \in P). \tag{3.2}$$

For the given map and separator, fix one choice of these points. The choices do not depend on the valuations subsequently considered or on the approximation parameter introduced below. All chosen points belong to $f[M]$. Different indices may determine the same point; when this occurs, the coefficients of the corresponding Dirac valuations are added.

Lemma 3.1. *The choices in* (3.2) *exist, and they satisfy*

$$\pi(x) \le u \implies f^2(x) \le a_u, \tag{3.3}$$

$$a \in u \implies a_u \le a. \tag{3.4}$$

Proof. Fix $u \in P$. Since $P = \pi[D]$, there is $y \in D$ with $\pi(y) = u$. By (2.1), choose $a \in M$ with $f(y) \leq a \leq y$. Then $a \in u$. For every $c \in u$, $f(y) \leq c$, so monotonicity of f gives

$$f(a) \leq f(y) \leq c.$$

Thus $f(a) \in f[u] \cap \bigcap_{c \in u} {\downarrow}c$, proving nonemptiness of the set in (3.2).

Write $a_u = f(m_u)$ for some $m_u \in u$. If $\pi(x) \leq u$, then $u \subseteq \pi(x)$ because P has the reverse-inclusion order. Hence $m_u \in \pi(x)$, which means $f(x) \leq m_u$. Applying f, we obtain $f^2(x) \leq f(m_u) = a_u$. Finally, (3.2) gives $a_u \leq a$ for every $a \in u$. □

For $\alpha = \sum_{x \in X} \alpha_x \delta_x \in \mathcal{B}(D)$, define

$$p^\alpha_u = \sum_{\substack{x \in X \\ \pi(x) = u}} \alpha_x \quad (u \in P), \qquad p^\alpha \in \Delta_{\leq 1}(P). \tag{3.5}$$

Consequently, for every $A \subseteq P$,

$$p^\alpha(A) = \sum_{\substack{x \in X \\ \pi(x) \in A}} \alpha_x, \qquad p^\alpha({\uparrow_P} u) = \sum_{\substack{x \in X \\ u \leq \pi(x)}} \alpha_x.$$

The coefficient vector in (3.5) records the total mass assigned to each element of the finite poset. This does not imply that the approximating maps factor through the subprobability powerdomain of that poset. If $\alpha \leq \beta$, put $X = \operatorname{supp}(\alpha) \cup \operatorname{supp}(\beta)$ and insert zero coefficients. For every $A \in \operatorname{Up}(P)$, the function $\mathbf{1}_A \circ \pi$ is monotone on X. Hence Lemma 2.4 gives

$$p^\alpha(A) = \sum_{x \in X} \alpha_x \mathbf{1}_A(\pi(x)) \leq \sum_{x \in X} \beta_x \mathbf{1}_A(\pi(x)) = p^\beta(A).$$

Therefore

$$\alpha \leq \beta \quad \Longrightarrow \quad p^\alpha \leq_{\mathrm{st}} p^\beta. \tag{3.6}$$

In particular, the mass assigned to each principal upper set of the finite poset cannot decrease when the input valuation increases. Throughout this section, the poset is determined by the fixed map and separator. The coefficient vectors of different input valuations are therefore defined on the same poset, even when their nonzero coordinates occur at different elements.

3.2 Monotone assignments indexed by subsets

Write $u \lessdot v$ when $u < v$ and there is no $w \in P$ with $u < w < v$. Thus $\lessdot$ is the cover relation of P. For $u \in P$, write $C(u) = \{v \in P : u \lessdot v\}$. Every strict comparison in P is a finite chain of cover relations. For $A \subseteq P$ and $u \in A$, let

$$U^A_u = \bigcup_{v \in C(u) \setminus A} {\uparrow_P} v. \tag{3.7}$$

The subset A is arbitrary; it is not assumed to be upper. Each U^A_u is upper in P. Define

$$\xi_A(x) = \mathbf{1}_A(\pi(x)) \delta_{f^2(x)} + \sum_{u \in A} \mathbf{1}_{U^A_u}(\pi(x)) \delta_{a_u}. \tag{3.8}$$

The value $\xi_A(x)$ is a positive simple valuation of mass at most $1 + |A|$, not necessarily a subprobability valuation.

Example 3.2. Let $D = \{0 < 1\}$, $f = \mathrm{id}_D$ and $M = D$. Identify P with D by $x \mapsto {\uparrow}x$. For $A = \{0\}$, keeping only the first term of (3.8) would give δ_0 at 0 and zero at 1, contrary to monotonicity. Here $U_0^A = \{1\}$ and $a_0 = 0$. The second term gives $\xi_A(0) = \xi_A(1) = \delta_0$.

Lemma 3.3. *For each $A \subseteq P$, the map $\xi_A : D \to \mathcal{V}(D)$ is monotone.*

Proof. Fix $A \subseteq P$ and let $x \leq y$. Since π is monotone and each U_u^A is upper in P,

$$\mathbf{1}_{U_u^A}(\pi(x)) \leq \mathbf{1}_{U_u^A}(\pi(y)) \qquad (u \in A).$$

For each $u \in A$, the displayed inequality $\mathbf{1}_{U_u^A}(\pi(x)) \leq \mathbf{1}_{U_u^A}(\pi(y))$ compares the term indexed by u in (3.8). If $\pi(x) \notin A$, the remaining term of $\xi_A(x)$ is zero, so these inequalities give $\xi_A(x) \leq \xi_A(y)$. If $\pi(x), \pi(y) \in A$, then $f^2(x) \leq f^2(y)$ and therefore $\delta_{f^2(x)} \leq \delta_{f^2(y)}$; adding this inequality to the inequalities for the terms indexed by A again gives $\xi_A(x) \leq \xi_A(y)$.

It remains to consider $u = \pi(x) \in A$ and $v = \pi(y) \notin A$. Since $u \leq v$ and $u \neq v$, choose a saturated chain

$$u = u_0 \lessdot u_1 \lessdot \cdots \lessdot u_r = v.$$

Let $j = \min\{a : u_a \notin A\}$, $w = u_{j-1}$, and $z = u_j$. Then $u \leq w \lessdot z \leq v$, $w \in A$, and $z \notin A$. If $u \in U_w^A$, then the definition of U_w^A gives an element $v' \in C(w) \setminus A$ with $v' \leq u$. Since $u \leq w$, this implies $v' \leq w$, whereas $v' \in C(w)$ implies $w < v'$. Antisymmetry would then give $v' = w$, contradicting $w < v'$. Hence $\pi(x) = u \notin U_w^A$. Since $z \notin A$ and $z \leq v$, $\pi(y) = v \in {\uparrow_P}z \subseteq U_w^A$.

Define $e : D \to \mathcal{V}(D)$ by

$$e(z') = \sum_{u' \in A \setminus \{w\}} \mathbf{1}_{U_{u'}^A}(\pi(z'))\delta_{a_{u'}} \qquad (z' \in D).$$

For every $u' \in A \setminus \{w\}$, the scalar-valued map $z' \mapsto \mathbf{1}_{U_{u'}^A}(\pi(z'))$ is monotone because π is monotone and $U_{u'}^A$ is upper in P. Hence the valuation-valued map $z' \mapsto \mathbf{1}_{U_{u'}^A}(\pi(z'))\delta_{a_{u'}}$ is monotone. Therefore their finite sum e is monotone, and $e(x) \leq e(y)$. Since $\pi(x) \leq w$, (3.3) gives $f^2(x) \leq a_w$. Consequently,

$$\xi_A(x) = \delta_{f^2(x)} + e(x) \leq \delta_{a_w} + e(y) = \xi_A(y).$$

□

3.3 Comparison of maps indexed by different subsets

Fix a positive real number t_u for each $u \in P$. Set

$$\tau_A = \sum_{u \in A} \left(\sum_{v \in C(u) \cap A} t_v \right) \delta_{a_u}, \tag{3.9}$$

and, for a simple valuation $\alpha = \sum_{x \in X} \alpha_x \delta_x$, define

$$T_A(\alpha) = \sum_{x \in X} \alpha_x \xi_A(x) + \tau_A \tag{3.10}$$

$$= \sum_{\substack{x \in X \\ \pi(x) \in A}} \alpha_x \delta_{f^2(x)} + \sum_{u \in A} \left(p^\alpha(U_u^A) + \sum_{v \in C(u) \cap A} t_v \right) \delta_{a_u}. \tag{3.11}$$

The map $T_A : \mathcal{B}(D) \to \mathcal{V}(D)$ may have values of mass greater than one. All operations in (3.10) are finite; the extension to nonsimple inputs will be performed only in section 4.

Lemma 3.4. *For fixed $A \subseteq P$, the map T_A is monotone on $\mathcal{B}(D)$.*

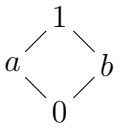
Proof. Let $\alpha \leq \beta$ and set $X = \operatorname{supp}(\alpha) \cup \operatorname{supp}(\beta)$. Write $\alpha = \sum_{x\in X} \alpha_x \delta_x$ and $\beta = \sum_{x\in X} \beta_x \delta_x$, inserting zero coefficients where necessary. Fix $U \in \sigma(D)$. By Lemma 3.3, $w_U(x) = \xi_A(x)(U)$ is a nonnegative monotone real-valued function on X. Thus Lemma 2.4 implies

$$T_A(\alpha)(U) = \sum_{x\in X} \alpha_x w_U(x) + \tau_A(U) \leq \sum_{x\in X} \beta_x w_U(x) + \tau_A(U) = T_A(\beta)(U).$$

Since U was arbitrary, $T_A(\alpha) \leq T_A(\beta)$. □

The inclusion $A \subseteq B$ does not in general imply $\xi_A(x) \leq \xi_B(x)$. The next lemma shows that the terms τ_A and τ_B provide a comparison between $T_A(\alpha)$ and $T_B(\alpha)$ under the explicitly stated inequalities on $p^\alpha(\uparrow_P v)$.

Example 3.5. Let $D = \{0, a, b, 1\}$ with $0 < a, b < 1$ and $a \parallel b$. Take $f = \mathrm{id}_D$ and $M = D$. Identify P with D by $x \mapsto \uparrow x$. The following Hasse diagram shows the cover relation.



For $A = \{0\}$,

$$U_0^A = \uparrow a \cup \uparrow b = \{a, b, 1\}, \qquad \xi_A(x) = \delta_0 \quad (x \in D).$$

The point 1 belongs to both principal upper sets, but its contribution to the term indexed by 0 is counted once.

Now take $A = \{0, a, b\}$ and $B = P$. Then

$$\xi_A(1) = \delta_a + \delta_b, \qquad \xi_B(1) = \delta_1.$$

Thus $A \subseteq B$ does not imply $\xi_A \leq \xi_B$. On the other hand,

$$\tau_A = (t_a + t_b)\delta_0, \qquad \tau_B = \tau_A + t_1(\delta_a + \delta_b).$$

If $0 \leq v \leq \min\{1, t_1\}$ and $\alpha = v\delta_1$, then

$$T_A(\alpha) = \tau_A + v(\delta_a + \delta_b) \leq \tau_B \leq T_B(\alpha).$$

Hence the terms τ_A and τ_B give $T_A(\alpha) \leq T_B(\alpha)$ although $\xi_A(1) \not\leq \xi_B(1)$.

Lemma 3.6. *Let $A \subseteq B \subseteq P$ and $\alpha \in \mathcal{B}(D)$. If*

$$p^\alpha(\uparrow_P v) \leq t_v \qquad (v \in B \setminus A),$$

then $T_A(\alpha) \leq T_B(\alpha)$.

Proof. For $u \in A$, put

$$Z_u = \bigcup_{v \in C(u) \cap (B\setminus A)} \uparrow_P v.$$

Since $P \setminus A = (P \setminus B) \cup (B \setminus A)$,

$$U_u^A = U_u^B \cup Z_u, \qquad \mathbf{1}_{U_u^A} \le \mathbf{1}_{U_u^B} + \mathbf{1}_{Z_u}.$$

Also $\mathbf{1}_A \le \mathbf{1}_B$. Hence, for every $x \in D$,

$$\xi_A(x) \le \xi_B(x) + \sum_{u \in A} \mathbf{1}_{Z_u}(\pi(x))\delta_{a_u}.$$

Write $\alpha = \sum_{x \in X} \alpha_x \delta_x$. Multiplication by α_x and finite summation give

$$\sum_{x \in X} \alpha_x \xi_A(x) \le \sum_{x \in X} \alpha_x \xi_B(x) + \sum_{u \in A} p^\alpha(Z_u)\delta_{a_u}.$$

For each $u \in A$, finite subadditivity and the hypothesis imply

$$p^\alpha(Z_u) \le \sum_{v \in C(u) \cap (B \setminus A)} p^\alpha(\uparrow_P v) \le \sum_{v \in C(u) \cap (B \setminus A)} t_v.$$

Define the positive simple valuation

$$\Delta = \sum_{u \in A} \sum_{v \in C(u) \cap (B \setminus A)} t_v \delta_{a_u}.$$

The definition of τ_B gives $\tau_A + \Delta \le \tau_B$. Therefore

$$T_A(\alpha) \le \sum_{x \in X} \alpha_x \xi_B(x) + \Delta + \tau_A \le \sum_{x \in X} \alpha_x \xi_B(x) + \tau_B = T_B(\alpha).$$

□

For a simple valuation α, set

$$A_\alpha = \{u \in P : p^\alpha(\uparrow_P u) > t_u\}, \qquad T(\alpha) = T_{A_\alpha}(\alpha). \tag{3.12}$$

If $\alpha \le \beta$, then (3.6) gives $A_\alpha \subseteq A_\beta$. The definition of A_α also gives $p^\alpha(\uparrow_P v) \le t_v$ for every $v \in A_\beta \setminus A_\alpha$. Lemma 3.6 therefore yields $T_{A_\alpha}(\alpha) \le T_{A_\beta}(\alpha)$, and Lemma 3.4 yields $T_{A_\beta}(\alpha) \le T_{A_\beta}(\beta)$. Hence

$$T(\alpha) = T_{A_\alpha}(\alpha) \le T_{A_\beta}(\alpha) \le T_{A_\beta}(\beta) = T(\beta). \tag{3.13}$$

Thus T is monotone for every family of positive real numbers $(t_u)_{u \in P}$.

3.4 Finite separation of the approximating maps

Let $\ell(u)$ be the largest number of elements in a chain of P ending at u, and put

$$H = \max_{u \in P} \ell(u), \qquad c_P = |\{(u, v) \in P^2 : u \lessdot v\}|.$$

A chain of P is a strictly decreasing chain of nonempty subsets of M, so $1 \le H \le \min\{|P|, |M|\}$. For an integer $m \ge 2$, choose

$$t_u = m^{-\ell(u)}, \qquad \lambda_m = \frac{m}{m + c_P + 1}, \qquad h_{f,M,m}(\alpha) = \lambda_m T(\alpha). \tag{3.14}$$

The chosen points a_u depend only on (f, M). We shall use

$$m^{-H} \le t_u \le m^{-1}, \qquad u < v \implies t_v \le t_u/m. \tag{3.15}$$

Proposition 3.7. *The map $h = h_{f,M,m}$ is a monotone map from $\mathcal{B}(D)$ to $\mathcal{B}(D)$. Define*

$$\begin{aligned} N &= |M|(m + c_P + 1)m^H + 1, \\ \Gamma_{f,M,m} &= \left\{ \sum_{a \in M} p_a \delta_a : p \in M_N(M) \right\}. \end{aligned} \tag{3.16}$$

Here M has the order induced from D, and $M_N(M)$ is the coefficient grid of (2.3)*. For every $\alpha \in \mathcal{B}(D)$, some $\gamma \in \Gamma_{f,M,m}$ satisfies*

$$h(\alpha) \leq \gamma \leq \alpha.$$

Proof. Equation (3.13) and $\lambda_m > 0$ imply that h is monotone. If $\alpha = 0$, then $p^\alpha(\uparrow_P u) = 0 < t_u$ for every u. Thus $A_\alpha = \varnothing$ and $h(0) = 0$. Taking $\gamma = 0 \in \Gamma_{f,M,m}$ gives $h(0) \leq \gamma \leq 0$. For the remainder of the proof, let α be nonzero and write $\alpha = \sum_{x \in X} \alpha_x \delta_x$, where $X = \operatorname{supp}(\alpha)$ is finite and $\alpha_x > 0$ for every $x \in X$.

For each $x \in X$, choose $c_x \in M$ with $f(x) \leq c_x \leq x$, and set

$$\theta = \theta_\alpha = \sum_{x \in X} \alpha_x \delta_{c_x} = \sum_{a \in M} \theta_a \delta_a, \qquad \theta_a = \sum_{\substack{x \in X \\ c_x = a}} \alpha_x. \tag{3.17}$$

Since $c_x \leq x$, we have $\delta_{c_x} \leq \delta_x$, and hence $\theta \leq \alpha$. The points c_x are used only to construct a valuation $\gamma \in \Gamma_{f,M,m}$ satisfying $h(\alpha) \leq \gamma \leq \alpha$; they do not occur in the definition of h.

For $u \in P$, the implication

$$u \leq \pi(x) \quad \Longrightarrow \quad c_x \in \pi(x) \subseteq u \quad \Longrightarrow \quad a_u \leq c_x$$

follows from (3.4). Therefore

$$p^\alpha(\uparrow_P u)\delta_{a_u} \leq \sum_{\substack{x \in X \\ u \leq \pi(x)}} \alpha_x \delta_{c_x} \leq \theta. \tag{3.18}$$

Put $A = A_\alpha$ and

$$B_u = p^\alpha(U_u^A) + \sum_{v \in C(u) \cap A} t_v \qquad (u \in A).$$

If $v \notin A$, then $p^\alpha(\uparrow_P v) \leq t_v$. Hence

$$\begin{aligned} B_u &\leq \sum_{\substack{u \lessdot v \\ v \notin A}} p^\alpha(\uparrow_P v) + \sum_{\substack{u \lessdot v \\ v \in A}} t_v \\ &\leq \sum_{u \lessdot v} t_v. \end{aligned} \tag{3.19}$$

If $u \in A$ and $u \lessdot v$, then $t_v \leq t_u/m < p^\alpha(\uparrow_P u)/m$. By (3.18), $t_v \delta_{a_u} \leq \theta/m$. Since at most c_P cover relations occur,

$$\sum_{u \in A} B_u \delta_{a_u} \leq \frac{c_P}{m} \theta.$$

Moreover, $f^2(x) \leq f(x) \leq c_x$, so

$$\sum_{\substack{x \in X \\ \pi(x) \in A}} \alpha_x \delta_{f^2(x)} \leq \theta.$$

Applying (3.11) yields

$$h(\alpha) \le \lambda_m(1 + c_P/m)\theta = \left(1 - \frac{1}{m + c_P + 1}\right)\theta. \tag{3.20}$$

Consequently, $h(\alpha) \le \alpha$ and $h(\alpha)(D) \le 1$. Since $h(\alpha)$ is a finite positive sum of Dirac valuations, $h(\alpha) \in \mathcal{B}(D)$.

Fix $U \in \sigma(D)$ and suppose $h(\alpha)(U) > 0$. By (3.11), either there is an $x \in X$ with $\pi(x) \in A$, $\alpha_x > 0$, and $f^2(x) \in U$, or there is a $u \in A$ with $B_u > 0$ and $a_u \in U$. In the first case, $f^2(x) \le a_{\pi(x)}$ and the upperness of U imply $a_{\pi(x)} \in U$. In the second case, $a_u \in U$ already holds. Thus in either case there is $u \in A$ such that $a_u \in U$. By (3.18) and (3.15),

$$\theta(U) \ge p^\alpha(\uparrow_P u) > t_u \ge m^{-H}. \tag{3.21}$$

Put $\varepsilon = m^{-H}/(m + c_P + 1)$. If $h(\alpha)(U) > 0$, then (3.20) and (3.21) give

$$h(\alpha)(U) \le \theta(U) - \frac{\theta(U)}{m + c_P + 1} < \theta(U) - \varepsilon.$$

If $h(\alpha)(U) = 0$, then $h(\alpha)(U) \le (\theta(U) - \varepsilon)^+$ because the right-hand side is nonnegative. Combining the two cases gives

$$h(\alpha)(U) \le \Big(\theta(U) - \varepsilon\Big)^+ \qquad (U \in \sigma(D)). \tag{3.22}$$

For each $a \in M$, set

$$\lambda_a^\theta = \frac{\lfloor N\theta_a \rfloor}{N}, \qquad \gamma = \sum_{a \in M} \lambda_a^\theta \delta_a.$$

Since $\sum_a \theta_a = \alpha(D) \le 1$,

$$\sum_{a \in M} \lfloor N\theta_a \rfloor \le N \sum_{a \in M} \theta_a \le N.$$

Thus $\lambda^\theta \in M_N(M)$ and $\gamma \in \Gamma_{f,M,m}$. Moreover, $\lambda_a^\theta \le \theta_a$ for all a, so $\gamma \le \theta \le \alpha$. For every $U \in \sigma(D)$,

$$0 \le \theta(U) - \gamma(U) = \sum_{a \in M \cap U} (\theta_a - \lambda_a^\theta) < \frac{|M|}{N} < \varepsilon.$$

If $\theta(U) > \varepsilon$, then $\gamma(U) > \theta(U) - \varepsilon$. If $\theta(U) \le \varepsilon$, then $\gamma(U) \ge 0$. Therefore (3.22) implies

$$h(\alpha)(U) \le \Big(\theta(U) - \varepsilon\Big)^+ \le \gamma(U) \qquad (U \in \sigma(D)).$$

Consequently $h(\alpha) \le \gamma \le \alpha$. □

For finite D with $f = \mathrm{id}_D$ and $M = D$, necessarily $c_x = x$, so $\theta_\alpha = \alpha$. Thus the proof of Proposition 3.7 reduces in this case to downward rounding of the original coefficient vector, as in [4, Proposition 3.13]. For general D, only the separating witness is moved to M; no monotonicity of $\alpha \mapsto \theta_\alpha$ is asserted.

Example 3.8. Return to $D = \{0 < 1 < 2\}$ with $f = \mathrm{id}_D$ and $M = D$. For $m = 2$,

$$(t_0, t_1, t_2) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\right), \qquad c_P = 2, \qquad \lambda_m = \frac{2}{5}.$$

Take $\alpha = \frac{1}{2}\delta_0 + \frac{1}{4}\delta_1 + \frac{1}{8}\delta_2$. Then

$$\left(p^\alpha(\uparrow_P 0), p^\alpha(\uparrow_P 1), p^\alpha(\uparrow_P 2)\right) = \left(\frac{7}{8}, \frac{3}{8}, \frac{1}{8}\right), \qquad A_\alpha = \{0, 1\}.$$

Since $p^\alpha(\uparrow_P 2) = t_2 = 1/8$ and A_α is defined by a strict inequality, $2 \notin A_\alpha$. The formulas give

$$\sum_x \alpha_x \xi_{A_\alpha}(x) = \frac{1}{2}\delta_0 + \frac{3}{8}\delta_1, \qquad \tau_{A_\alpha} = \frac{1}{4}\delta_0,$$

so

$$T(\alpha) = \frac{3}{4}\delta_0 + \frac{3}{8}\delta_1, \qquad h(\alpha) = \frac{3}{10}\delta_0 + \frac{3}{20}\delta_1.$$

The auxiliary valuation $T(\alpha)$ has mass $9/8$. Multiplication by the fixed number λ_m gives $h(\alpha)(D) \le 1$; no normalization depending on α is used. Here $N = 121$ in (3.16), and rounding $\theta = \alpha$ gives the witness

$$\gamma = \frac{60}{121}\delta_0 + \frac{30}{121}\delta_1 + \frac{15}{121}\delta_2, \qquad h(\alpha) \le \gamma \le \alpha.$$

Example 3.9. Consider the diamond of Example 3.5 with $m = 2$. Then

$$t_0 = \frac{1}{2}, \qquad t_a = t_b = \frac{1}{4}, \qquad t_1 = \frac{1}{8}.$$

For $s, t \in \{\frac{1}{8}, \frac{1}{4}\}$, put

$$\alpha_{s,t} = \left(\frac{5}{8} - s - t\right)\delta_0 + s\delta_a + t\delta_b + \frac{1}{8}\delta_1.$$

All four inputs have mass $3/4$. Write $r_u = p^{\alpha_{s,t}}(\uparrow u)$. Then $r_0 = 3/4$, $r_1 = 1/8$, $r_a = s + 1/8$, and $r_b = t + 1/8$. In (3.11), the coefficient added at 0 is

$$B_0(\alpha_{s,t}) = p^{\alpha_{s,t}}(U_0^{A_{\alpha_{s,t}}}) + \sum_{v \in \{a,b\} \cap A_{\alpha_{s,t}}} t_v.$$

If $a, b \notin A_{\alpha_{s,t}}$, then $p^{\alpha_{s,t}}(\uparrow a \cup \uparrow b) = s + t + 1/8$. If exactly one of a, b belongs to $A_{\alpha_{s,t}}$, then $B_0(\alpha_{s,t}) = 1/4 + 1/4$. If both belong to $A_{\alpha_{s,t}}$, then the same equality holds. Thus

$B_0(\alpha_{s,t})$	$t = 1/8$	$t = 1/4$
$s = 1/8$	$3/8$	$1/2$
$s = 1/4$	$1/2$	$1/2$

and

$$B_0(\alpha_{1/4,1/4}) - B_0(\alpha_{1/4,1/8}) - B_0(\alpha_{1/8,1/4}) + B_0(\alpha_{1/8,1/8}) = -\frac{1}{8}.$$

Since r_0, r_1 are fixed and r_a, r_b vary independently, the coefficient $B_0(\alpha_{s,t})$ cannot be written as $F(r_a) + G(r_b) + c$ on the four inputs. Because $1 \in \uparrow a \cap \uparrow b$, the value $p^{\alpha_{s,t}}(\uparrow a \cup \uparrow b)$ cannot be decomposed into one term depending only on r_a and another depending only on r_b on these four inputs. The nonzero mixed difference displayed above concerns the basis map $h_{f,M,2}$, not the self-composition $\widehat{h}_{f,M,2} \circ \widehat{h}_{f,M,2}$ used in Theorem 4.3.

Remark 3.10. Only the c_P cover relations occur in the coefficients added in (3.9). For an n-element chain there are $n-1$ cover relations and $n(n-1)/2$ strict comparable pairs. Using only cover relations does not change the maximum chain length H and gives the smaller value $c_P = n-1$ in λ_m and in the sufficient grid size N from (3.16). We do not claim a pointwise comparison between $h_{f,M,m}$ and the map obtained by replacing each set $C(u)$ with $\{v \in P : u < v\}$ in (3.7) and (3.9).

4 Basis extension and the FS approximate identity

This section extends the maps constructed on simple valuations to the entire subprobability powerdomain. We first show that the extensions are Scott continuous and retain finite separators. We then prove that one of these maps can approximate any prescribed finite family of way-below pairs. Combining these results with the basis criterion establishes the FS-domain property of the subprobability powerdomain.

Assume now that D is continuous. The simple valuations form a basis of $E = \mathcal{V}_{\le 1}(D)$ by Lemma 2.3. For fixed f, M, m, define

$$\widehat{h}_{f,M,m}(\nu) = \sup_{\substack{\alpha \in \mathcal{B}(D) \\ \alpha \ll \nu}} h_{f,M,m}(\alpha). \tag{4.1}$$

By Proposition 3.7 and Lemma 2.5, formula (4.1) defines a Scott-continuous self-map of E, and $\Gamma_{f,M,m}$ separates $\widehat{h}_{f,M,m}$ from id_E. To apply Proposition 2.6, it remains to verify condition (ii) of that proposition on the basis $\mathcal{B}(D)$.

4.1 Simultaneous approximation of finitely many simple valuations

Lemma 4.1. *Fix f, M. Given simple valuations $\alpha_1, \dots, \alpha_r$ and $0 < c < 1$, all sufficiently large integers m satisfy*

$$c \sum_{x \in \mathrm{supp}(\alpha_j)} (\alpha_j)_x \delta_{f^2(x)} \le h_{f,M,m}(\alpha_j) \qquad (1 \le j \le r).$$

Proof. Write $\alpha_j = \sum_{x \in X_j} a_{j,x} \delta_x$, where $X_j = \mathrm{supp}(\alpha_j)$ and all displayed coefficients are positive. If $\alpha_j = 0$, take $X_j = \varnothing$. Define

$$\rho = \min\Big(\{1\} \cup \{a_{j,x} : 1 \le j \le r,\ x \in X_j\}\Big) > 0.$$

Choose an integer

$$m > \max\left\{1, \rho^{-1}, \frac{c(c_P+1)}{1-c}\right\}.$$

Then $m \ge 2$, $1/m < \rho$, and $\lambda_m > c$. If $x \in X_j$, then

$$p^{\alpha_j}(\uparrow_P \pi(x)) \ge a_{j,x} \ge \rho > 1/m \ge t_{\pi(x)}.$$

Hence $\pi(x) \in A_{\alpha_j}$ for every $x \in X_j$. Since all other coefficients in (3.11) are nonnegative,

$$h_{f,M,m}(\alpha_j) \ge \lambda_m \sum_{x \in X_j} a_{j,x} \delta_{f^2(x)} \ge c \sum_{x \in X_j} a_{j,x} \delta_{f^2(x)}.$$

If $X_j = \varnothing$, both sides of the asserted inequality are zero, because $h_{f,M,m}(0) = 0$. Every larger integer m still satisfies $m \ge 2$, $1/m < \rho$, and $\lambda_m > c$. □

Lemma 4.2. *Let $(f_i)_{i\in I}$ be an FS approximate identity on D. For each i, choose a finite separator M_i and make the choices in (3.2) for the finite poset $P_i = \pi_{f_i,M_i}[D]$. Given finitely many $\alpha_j \in \mathcal{B}(D)$ and $u_j \ll \alpha_j$ in $\mathcal{V}_{\le 1}(D)$, there are common indices $i \in I$ and $m \ge 2$ such that*

$$u_j \le h_{f_i,M_i,m}(\alpha_j) \qquad \text{for every } j.$$

Proof. If the list is empty, choose any $i \in I$ and $m \ge 2$. Otherwise write $\alpha_j = \sum_{x\in X_j} a_{j,x}\delta_x$ for $1 \le j \le r$, where $X_j = \operatorname{supp}(\alpha_j)$ and the displayed coefficients are positive.

If $i \le i'$, then $f_i^2 \le f_{i'}^2$. For $i, k \in I$, choose $\ell \ge i, k$; then $f_i(f_k(x)) \le f_\ell^2(x)$ for every $x \in D$. Thus the subset $\{(\ell,\ell) : \ell \in I\}$ is cofinal in $I \times I$ with its componentwise order. Scott continuity gives

$$\sup_i f_i^2(x) = \sup_{i,k} f_i(f_k(x)) = \sup_i f_i\left(\sup_k f_k(x)\right) = x. \tag{4.2}$$

For $i \in I$ and $0 < s < 1$, set

$$\beta_{i,s,j} = s \sum_{x\in X_j} a_{j,x}\delta_{f_i^2(x)}.$$

Order $I \times (0,1)$ componentwise, using the usual order on $(0,1)$. For each j, the family $(\beta_{i,s,j})_{(i,s)\in I\times(0,1)}$ is directed. Indeed, $i \le i'$ implies $f_i^2(x) \le f_{i'}^2(x)$ for every x, and $s \le s'$ implies $sa_{j,x} \le s'a_{j,x}$ for every $x \in X_j$. Moreover, $f_i^2(x) \le x$ and $s < 1$ imply $\beta_{i,s,j} \le \alpha_j$. Fix $U \in \sigma(D)$. For every $x \in X_j \cap U$, (4.2) and Scott openness give an index i_x with $f_{i_x}^2(x) \in U$. Since $X_j \cap U$ is finite, there is an index $i_U \in I$ above all i_x. Hence $f_i^2(x) \in U$ for every $x \in X_j \cap U$ and every $i \ge i_U$. For $x \in X_j \setminus U$, the inequalities $f_i^2(x) \le x$ imply $f_i^2(x) \notin U$. Consequently,

$$\beta_{i,s,j}(U) = s\alpha_j(U) \qquad (i \ge i_U).$$

If $X_j \cap U = \varnothing$, this equality holds at every index. Since $\sup_{0<s<1} s\alpha_j(U) = \alpha_j(U)$,

$$\alpha_j = \sup_{i\in I,\, 0<s<1} \beta_{i,s,j}. \tag{4.3}$$

Since $u_j \ll \alpha_j$, choose $i_j \in I$ and $0 < s_j < 1$ with $u_j \le \beta_{i_j,s_j,j}$. Choose $i \ge i_1, \dots, i_r$ and $s = \max_j s_j < 1$. Then $u_j \le \beta_{i,s,j}$ for every j. With this i fixed, Lemma 4.1 supplies one integer $m \ge 2$ such that

$$u_j \le \beta_{i,s,j} \le h_{f_i,M_i,m}(\alpha_j) \qquad (1 \le j \le r).$$

□

4.2 Proof of the closure theorem

Theorem 4.3. *If D is an FS-domain, then $\mathcal{V}_{\leq 1}(D)$ is a pointed FS-domain.*

Proof. Let $(f_i)_{i\in I}$ be an FS approximate identity on D. For each i, choose a finite separator M_i and make the choices in (3.2) for the finite poset $P_i = \pi_{f_i,M_i}[D]$. By Lemmas 2.2 and 2.3, $E = \mathcal{V}_{\leq 1}(D)$ is a continuous dcpo with basis $B = \mathcal{B}(D)$. Set

$$\mathcal{A} = \{h_{f_i,M_i,m} : i \in I,\ m \geq 2\}.$$

For each $h = h_{f_i,M_i,m}$, Proposition 3.7 gives monotonicity and a fixed finite separator $\Gamma_{f_i,M_i,m}$ on B. By Lemma 4.2, the family $\mathcal{A}$ satisfies condition (ii) of Proposition 2.6. Write $\widehat{h}_{i,m} = \widehat{h}_{f_i,M_i,m}$ for the extensions in (4.1). Proposition 2.6 therefore gives the FS approximate identity

$$\mathcal{G} = \{\widehat{h}_{i,m} \circ \widehat{h}_{i,m} : i \in I,\ m \geq 2\}$$

on E. Finally, $0(U) = 0$ for every $U \in \sigma(D)$ defines a continuous subprobability valuation, and $0 \leq \nu$ for all $\nu \in E$. Thus E is pointed, whether or not D has a least element. □

In particular, the full subcategory of pointed FS-domains is closed under the subprobability powerdomain.

Remark 4.4. The family $\mathcal{G}$ is ordered pointwise. Its directedness does not assert monotonicity of $m \mapsto \widehat{h}_{i,m}$ or compatibility between different finite separators M_i. For $\alpha = \sum_x \alpha_x \delta_x$, the value $h_{f_i,M_i,m}(\alpha)$ contains terms supported at $f_i^2(x)$. As α varies, the points x range over D, and the proof gives no finite subset of D containing all points $f_i^2(x)$ that can occur. Consequently, Proposition 3.7 does not produce a factorization through one fixed valuation domain $\mathcal{V}_{\leq 1}(P)$ with P finite.

Remark 4.5. For finite D, the extension (4.1) agrees with the basis map h. Indeed, every valuation on D is simple. To verify that $c\alpha \ll \alpha$ for $0 < c < 1$, let (ν_d) be directed and satisfy $\alpha \leq \sup_d \nu_d$. For each upper set U with $\alpha(U) > 0$, choose d_U such that $c\alpha(U) \leq \nu_{d_U}(U)$. There are only finitely many upper sets of D. Choose an index d above all the corresponding indices d_U; then $c\alpha(U) \leq \nu_d(U)$ for every upper set U, and hence $c\alpha \leq \nu_d$.

For fixed α, the set $A_{c\alpha}$ equals A_α for all $c < 1$ sufficiently close to 1. If $A = A_\alpha$, then

$$h(c\alpha) = \lambda_m\left(c\sum_x \alpha_x \xi_A(x) + \tau_A\right) \quad \text{for all such } c.$$

The valuations $h(c\alpha)$ increase with c, and the displayed formula converges pointwise on Scott-open sets to $h(\alpha)$. Therefore

$$h(\alpha) = \sup_{0<c<1} h(c\alpha) \leq \widehat{h}(\alpha) \leq h(\alpha).$$

The inequality $\widehat{h}(\alpha) \leq h(\alpha)$ follows from monotonicity of h and $\beta \ll \alpha \Rightarrow \beta \leq \alpha$. Thus Examples 3.8 and 3.9 also describe the Scott-continuous extensions $\widehat{h}$ before the self-composition $\widehat{h} \circ \widehat{h}$ is taken.

For a singleton $D = \{*\}$, the definition of h_m becomes

$$h_m(r\delta_*) = \frac{m}{m+1} r\mathbf{1}_{\{r>1/m\}}\delta_*.$$

The values at $(c/m)\delta_*$ for $0 < c < 1$, and at their supremum $(1/m)\delta_*$, are all zero. If $>$ were replaced by $\geq$, the modified map would remain zero at every $(c/m)\delta_*$ but would be positive at their supremum. The modified map would therefore fail to preserve this directed supremum and would not be Scott continuous. Its basis extension would still have value zero at $(1/m)\delta_*$.

The dcpo D need not be pointed. For the two-element antichain $D = \{a, b\}$, the identity has separator $M = D$. Here P is an antichain, $H = 1$, and $c_P = 0$, so

$$h_m(u\delta_a + v\delta_b) = \frac{m}{m+1}\left(u\mathbf{1}_{\{u>1/m\}}\delta_a + v\mathbf{1}_{\{v>1/m\}}\delta_b\right), \qquad u, v \geq 0, \quad u + v \leq 1.$$

The coefficients removed by the indicator functions and by the factor $m/(m+1)$ are not transferred to any point of D. If empty dcpos are allowed, then $\mathcal{V}_{\leq 1}(\varnothing) = \{0\}$, which is an FS-domain with identity map separated by $\{0\}$.

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